



FORTEST

catch the leak.

Communication Protocol for MSeries and TSeries devices

Summary

1. Introduction.....	4
• 2. Direct Communication with PC Protocol Description	5
3.1 Command “1”: Instantaneous Status Request of the Instrument	6
3.2 Command “2”: Read Test Result (Result Stack)	9
3.3 Command “3”: Read Instrument Version	12
3.4 Command “4”: Reset and/or Piece Counter Reading only	14
3.5 Command “5”: Write the Program Number	15
3.6 Command “6”: Special Functions like Keys.....	16
3.7 Command “7”: Command "7": Read Barcode Command (optional)	17
3.8 Command “8”:	18
3.9 Command “9”:	19
3.10 Command “A”:	20
3.11 Command “B”: Read a Menu Parameter	21
3.12 Command “C”: Write a Parameter of a menu.....	23
3.13 Command “D”:	24
3.14 Command “E”: Write a Memory Location	25
3.15 Command “F”: Set Date and Time	26
3.17 Command “G”:	27
3.17 Command “H”:	28
3.18 Command “I”:	29
3.19 Command “J”:	30
3.20 Command “K”:	31
3.21 Command “L”: Read Sensors	32
3.22 Command “M”: Calibrate	33
3.23 Command “N”: Read Calibration Points	34
4. Protocol Communication Through TTY Description	35
5. Test End Track Operation Description	36
Appendix 1: Checksum Calculation	37
Appendix 2: Coding of the Test Phases	38
Appendix 3: Coding of Results.....	39
Appendix 4: Index Coding and Tables for Reading and Writing Parameters	40
Appendix 5: Coding of the Unit of Measure.....	47
Appendix 6: Coding of the I/O Masks	48

Appendix 7: Memories and Mapping of the same	50
Appendix 8 : Linked tests codification	51
Changelog	52

1. Introduction

Different choices of communication are possible on every COM port of the instrument, depending on the relative parameter being set, or **"COM1:"** / **"COM1 Mode:"**. Remember that, depending on the parameter value, they are:

- if parameter is worth **"0"**, no communication is active
- if parameter is worth **"1"**, direct communication with PC (or other device with a serial port) is active: the instrument only responds if PC asks something, in a way PC acts as master and the instrument as slave. See paragraph 1 for the communication protocol
- if parameter is worth **"2"**, communication with PC (or other device with a serial port) is active, but through TTY (ring current): the instrument behaves as in case "1". See paragraph 2 for the communication protocol
- if parameter is worth **"3"**, the test end track is active: at the end of each test (good, reject or abort), the instrument spontaneously sends a string with significant data of the just completed test: there is no possibility of re-transmission, as the instrument does not respond to PC requests. See paragraph 3 for the communication protocol

Attention; the above is only valid for the COM1 serial.

Depending on the instrument used for communication, some communication speeds (baud rate) may be more or less sensitive to interferences, communication cable lengths, etc. Therefore, below is a list of recommended speeds depending on the instrument:

- M2xxx: 9600bps, 19200bps.
- M7xxx: 4800bps, 9600bps, 19200bps.
- M6xxx, M8xxx and M9xxx with PLL off: 4800bps, 9600bps, 19200bps, 38400bps.
- M6xxx, M8xxx e M9xxx with PLL on: 2400bps, 19200bps, 38400bps, 57600bps, 115200bps.

2. Direct Communication with PC Protocol Description

Communication with the instrument happens with the following default settings:

- bps speed: set by means of relative setup parameter
- data bit: 8
- equality: none
- stop bit: one.
- flow control: none.

ASCII characters are sent: with regard to the format, all addresses are passed in hexadecimal format, data in decimal (unless otherwise indicated), and both converted in ASCII characters.

As shown below, second and third character of the communication strings between instrument and PC are dedicated to the serial address of the instrument. This address must be inserted in hexadecimal format (e.g. address 1 will be inserted in the string as "01", address 30 will be inserted as "1E").

Check the serial address set in the setup menu of the instrument is consistent with that inserted in the communication commands request strings between the PC and the instrument.

A series of controls are implemented inside the instrument to prevent incorrect contents of the command strings sent by the user (incorrect addresses, data outside the admitted limits, functions requested when they cannot be executed, etc.): however, it is advised to carefully read this document being the instrument without human intelligence, it cannot compensate logic-type errors. For example, with regard to the writing of data, both those to be written in EEPROM and those in RAM, they are limited between 0-65535.

The general syntax of the sending of a command by the PC (or another master device in communication) envisions:

- an opening character, ":"
- two characters indicating the serial address of the machine
- a character indicating the command to be executed
- various or no characters which number and meaning depends on the type of command, and for the meaning of which reference is made in the successive paragraphs
- two checksum characters (to calculate checksum see Appendix 1).

The general response syntax of the instrument is exactly equal to that of the command.

The instrument responds if, and only if, the received serial address coincides with its own and if the received checksum is correct. The instrument is not enabled to request re-transmission in case of incorrect checksum, it works only as slave.

Below, to formulate examples of messages to be transmitted and received, it will be assumed that the serial address of the instrument is 1, therefore "01" will be the value of the corresponding field.

WARNING: ensure to pay attention to the writing in memories, to avoid writing data in unwanted positions and not remembering to have done so.

3.1 Command "1": Instantaneous Status Request of the Instrument

With regard to the command, the command character is "1" and there are no data characters. Therefore, the message is:

":0116D".

With regard to the response, therefore starting with ":011" and ending with two checksum characters (to calculate checksum see Appendix 1), it contains:

Section with a range of fields to understand the status the instrument is currently in

- four characters that code the status of the errors active in the instrument. A number in exadecimal format is transmitted, representing a number in hexadecimal format in which, at every bit, a type of error is associated. See this correspondence below.
- two characters indicating the status of the instrument. See below for the transmitted status and value correspondence.
- two characters indicating the substatus of the instrument. See below for the transmitted substatus and value correspondence, also depending on the status
- two characters indicating the currently active result in the instrument. See Appendix 3 for the number/meaning association
- two characters indicating an auxiliary status, not currently used and, therefore, at zero
- five characters indicating the program number currently active in the instrument.
- five characters indicating the number of memorised and not yet read results (see command "2")

Section with information relating to the last modified parameter, information that can be used to update the parameters memory on connected PC

- two characters indicating the menu containing the last modified parameter (see Appendix 4 for the number/meaning association)
- three characters indicating the parameter index in the above menu (the last modified parameter) (see Appendix 4 for the number/meaning association)
- two characters indicating the submenu containing the last modified parameter (see Appendix 4 for the number/meaning association)
- three characters indicating the parameter index in the above submenu (the last modified parameter) (see Appendix 4 for the number/meaning association)

Section relating to information on a time

- Ten characters indicating a time remaining to the end of a phase (substatus). Which phase, and of what, depends on some previous characters, or those indicating the status and substatus. If, for example, a test is being executed while in adjustment phase, this time indicates the remaining time to the end of the settling phase, and so on. For some status, this field may not have a meaning and be at zero.
- two characters containing the unit of measure associated to the above time (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above time (if, for example, the time field contains the value "112", the previous field refers to seconds and this field contains the value "2", meaning the value must be interpreted as 1.12s)

Section relating to the pressure measured by the instrument

- one character containing the mark of the measured pressure with code "0" = positive pressure, "1" = negative pressure
- ten characters containing the value of the measured pressure
- two characters containing the unit of measure associated to the above pressure (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above pressure

Section relating to the VOUT measured by the instrument

- one character containing the mark of the measured VOUT with code "0" = positive VOUT , "1" = negative VOUT
- ten characters containing the value of the measured VOUT
- two characters containing the unit of measure associated to the above VOUT (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above VOUT

Section relating to the temperature measured by the instrument

- one character containing the mark of the measured temperature with code "0" = positive temperature, "1" = negative temperature
- five characters containing the value of the measured temperature
- two characters containing the unit of measure associated to the above temperature (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above temperature

Section relating to the I/O of the instrument

- three characters containing the status of the eight inputs of the instrument (see Appendix 6 for the meaning)
- three characters containing the status of the eight inputs of the instrument (see Appendix 6 for the meaning)
- three characters containing the status of the four inputs and four outputs of the expansion board of the instrument (see Appendix 6 for the meaning)

With regard to the two characters for coding the errors active in the instrument, a number is received in the string in decimal format, which must be converted in hexadecimal format, in which an error is associated to each bit, or:

- bit0 to 1 indicates the FLASH error is active: should this error be active, immediately contact the after-sales assistance
- bit1 to 1 indicates the EEPROM error is active: should this error be active, access the test menu, modify a parameter, exit the test menu, check this bit again: should the error persist, immediately contact the after-sales assistance
- bit2 to 1 indicates the pressure high full scale error is active: should this error be active, switch-off the air delivery supply of the instrument, disconnect the test piece
- bit3 to 1 indicates the VOUT high full scale error is active: should this error be active, switch-off the air delivery supply of the instrument, disconnect the test piece
- bit4 to 1 indicates the outputs error is active: should this error be active, disconnect everything connected to the electric connectors of the instrument
- bit5 to 0 indicates the piece counter error is active: should this error be active, reset the piece counter (as explained in the manual): should the error persist, immediately contact the after-sales assistance
- bit6 to 1 indicates temperature high full scale error: should this error be active, it is possible there is an anomaly in the reading of the temperature. Check the temperature probe. Obviously, this error may occur if the optional "Temperature measurement" is present
- bit7 to 1 indicates error in the battery voltage measurement: present only in instruments M27xx, if active this bit indicates that the battery is discharged. If necessary, recharge the battery with the provided battery-charger. Once recharged, check this bit again: should the error persist, immediately contact the after-sales assistance
- bit8 to 1 for internal use; it is not required to check its value
- bit9 to 1 indicates an error relating to the electronic regulator: should this error arise, check there is air in the "IN AIR" fitting. Should the error persist, immediately contact the after-sales assistance
- bit10 to 1 for internal use; it is not required to check its value

With regard to the two characters identifying the status of the instrument, the correspondence between number and status is the following:

- "0" indicates no particular status active; instrument in stand-by
- "1" indicates the test status in progress as active
- "2" indicates the auto zero status in progress as active
- "3" indicates the discharged status in progress as active
- "4" indicates the calibration status of the calibrated leak for the bell test in progress as active
- "5" indicates the lids status in execution as active

With regard to the two characters identifying the substatus relating to the status of the instrument, the correspondence between number and substatus varies depending on the status: meaning that the same number can have a different meaning depending on the above status.

In case status "0" is active, no substatus can be active.

In case status "1" is active, refer to Appendix 2.

In case status "2" is active:

- "1" indicates the substatus of auto zero of the pressure in progress as active
- "2" indicates the substatus of auto zero of the VOUT in progress as active
- "3" indicates the substatus of auto zero error of the pressure as active
- "4" indicates the substatus of auto zero error of the VOUT as active

In case status "3" is active:

- "1" indicates the discharged substatus in progress as active
- "2" indicates the discharged substatus in progress as active
- "3" indicates the discharged substatus in progress as active
- "4" indicates the discharged substatus in progress as active

In case status "4" is active:

- "1" indicates the calibration substatus of the calibrated leak for the bell test in progress as active
- "2" indicates the calibration substatus of the calibrated leak for the bell test in progress as active

In case status "5" is active:

- "2" indicates the pre-test lids execution substatus as active (to understand which lid is in progress, refer to the fields indicating the status of the I/O of the instrument, Appendix 6 and the lids I/O association supplied with the instrument)
- "3" indicates the after test lids execution substatus as active (to understand which lid is in progress, refer to the fields indicating the status of the I/O of the instrument, Appendix 6 and the lids I/O association supplied with the instrument)

3.2 Command "2": Read Test Result (Result Stack)

With regard to the command, the command character is "2", follows a subcommand "s" made of two characters and the checksum. "s" can have the following values:

- s = "00" : the instrument will always and only send the last test result. E.g.: ":012000C"
- s = "01" : implements the stack of the results; the instrument deletes the last just read result. E.g.: ":012010B"

For example, should there be 3 outstanding results, a possible combination of the above subcommands could be:

s = "01" -> The last executed test is read, test3, and goes back by one position

s = "01" -> Test2 is read and goes back by one position

s = "01" -> Test1 is read and goes back by one position

After then, subsequent requests with s = "01" will give an error message, whereas subsequent requests with s = "00" will give the last executed result. For further information on the above and, in particular, on the operation of the stack of results, see OPERATION OF THE STACK OF RESULTS paragraph.

The instrument is able to memorise a given number of results: after explanation of the response fields, the operation is described.

With regard to the response, therefore starting with ":012xx" and ending with checksum (to calculate checksum see Appendix 1), it contains:

- five characters indicating the number of lost results. See the OPERATION OF THE STACK OF RESULTS paragraph to better understand the meaning of these characters
- five characters indicating the number of results not yet transmitted by the instrument to PC (the instrument considers the current result has just been read, therefore indicates the number of results that are still to be read). . See the OPERATION OF THE STACK OF RESULTS paragraph to better understand the meaning of these characters

Section relating to the moment the test has been completed, useful for chronological sort of the same tests during storing

- two characters indicating the time the test was completed (from "00" to "23")
- two characters indicating the minutes the test was completed
- two characters indicating the seconds the test was completed
- two characters indicating the day the test was completed
- two characters indicating the month the test was completed (from "01" for January to "12" for December)
- two characters indicating the year the test was completed (if the year is 2009, it will be "09")

Section of general information on the test

- five characters indicating the program number relating to the test of this result
- three characters indicating whether the test was linked or not
- three characters indicating the type of memorised test (see Appendix 4, TEST MENU table, test type)
- two characters indicating the test result (see Appendix 3 for the transmitted number/meaning association)
- two characters indicating the status the test ended in (see Appendix 2 for the meaning)

Section relating to information on the time

- ten characters indicating the time remaining to the end of the test phase
- two characters containing the unit of measure associated to the above time (see Appendix 5 for the

- number/meaning association)
- two characters indicating the number of decimal points associated to the above time

Section relating to the pressure memorised by the instrument relating to the test

- one character containing the mark of the memorised test pressure, with code "0" = positive pressure or pressure not yet measured (if the test has ended before the start of pressure measurement), "1" = negative pressure
- ten characters indicating the numerical value of the memorised pressure
- two characters containing the unit of measure associated to the above pressure (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above pressure

Section relating to the VOUT memorised by the instrument relating to the test

- one character containing the mark of the memorised VOUT pressure, with code "0" = positive VOUT or VOUT not yet measured (if the test has ended before the start of VOUT measurement), "1" = negative VOUT
- ten characters indicating the numerical value of the memorised VOUT
- two characters containing the unit of measure associated to the above VOUT (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above VOUT

Section relating to the VOUTAUX1 memorised by the instrument relating to the test

- one character containing the mark of the memorised VOUTAUX1 pressure, with code "0" = positive VOUTAUX1 or VOUTAUX1 not yet measured (if the test has ended before the start of VOUTAUX1 measurement), "1" = negative VOUTAUX1
- ten characters indicating the numerical value of the memorised VOUTAUX1
- two characters containing the unit of measure associated to the above VOUTAUX1 (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above VOUTAUX1

Section relating to the VOUTAUX2 memorised by the instrument relating to the test

- one character containing the mark of the memorised VOUTAUX2 pressure, with code "0" = positive VOUTAUX2 or VOUTAUX2 not yet measured (if the test has ended before the start of VOUTAUX2 measurement), "1" = negative VOUTAUX2
- ten characters indicating the numerical value of the memorised VOUTAUX2
- two characters containing the unit of measure associated to the above VOUTAUX2 (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above VOUTAUX2

Section relating to the temperature memorised by the instrument relating to the test

- one character containing the mark of the memorised test temperature with code "0" = positive temperature, "1" = negative temperature
- five characters indicating the numerical value of the memorised temperature
- two characters containing the unit of measure associated to the above temperature (see Appendix 5 for the number/meaning association)
- two characters indicating the number of decimal points associated to the above temperature

OPERATION OF THE STACK OF RESULTS

As said, the instrument is able to memorise a given number of results (to know how many, see instrument

version read command); these results are memorised with a LIFO (Last In First Out) type stack logic. Therefore, when a read result command is sent, the instrument ALWAYS responds with the last memorised result.

As explained above, there are 2 commands for reading a test result. Both always respond with the last memorised result. The difference between the 2 commands is that

- Command s = "01" allows reading the last result but deletes it from the list; useful for reading the lost results. This command can be sent as many times as are the memorised results; once all memorised results are finished (meaning the field identifying the remaining results is equal to 0 - see above), the instrument will respond with an error message containing letters "e"
- Command s = "00" allows reading always and only the last result; useful if wanting to re-read the last memorised result. This command can be sent as often as wanted.

The following example may be useful to better understand the operation of the stack of results:

Let us suppose that the instrument has executed three tests from last read result request, the last executed has given reject, the penultimate good, the last but two abort (the one with abort will be terminate at a time before that of good, which will terminate at a time prior to that of reject). If in this case a read result request is sent with subcommand s = "01", the instrument will respond with data of the test resulting in reject, and it will indicate there are still two unread results (outstanding results). Therefore, another read result request with subcommand s = "01" can be sent (without the instrument having executed other tests from last, meaning that resulting in reject), to which the instrument will respond with data relating to the test giving good result, and where it will indicate there is still one result to be read. If another read result request is sent with subcommand s = "01", the instrument will respond with data of the test resulting in abort, and it will indicate there are no unread results. If other read result requests are sent but with subcommand s = "00", the instrument will continue responding with data of the test resulting in abort, continuing to indicate there are no unread results. The strategy is that of a LIFO (Last In First Out) managed stack of results.

In responding to the read result command, the instrument supplies the number of results not yet read (and decreases this number each time it receives this command if the subcommand s is "01", to "make way" for the next results), and also supplies this in the response to the read status command (but this time without decreasing the number of results not yet read). For as long as the unread results are below the maximum number, it is always possible to request them using this command, and obtain them. If this limit is exceeded, the new result will ensure that the eldest will be lost.

However, the field identifying the number of lost results will be increased by 1 each time a result is lost, in order to inform the user that the maximum limit of memorised results has been exceeded.

All results are lost upon machine switch-off, the buffer reset, as is the timer of outstanding results.

3.3 Command “3”: Read Instrument Version

With regard to the command, the command character is “3” and no data characters are requested: therefore, the message is:

“:0136B”.

With regard to the response, it contains:

- ten characters indicating the serial number of the instrument
- four characters indicating the total checksum in hexadecimal format of the FW version
- four characters indicating the boot checksum in hexadecimal format of the FW version

Section with information to identify the instrument code

- five characters indicating the instrument type (the first five characters of the code)
- a signalling character “-“
- three characters indicating the high full scale in direct pressure of the instrument
- three characters indicating the high full scale in VOUT of the instrument
- a signalling character “-“
- three characters indicating the power supply voltage, fittings and type of gas of the instrument, as from code of the same (for correspondence between number and meaning, refer to the coding of the ForTest instruments)
- four characters that, in hexadecimal format, indicate the active pneumatic optionals (for correspondence between number and meaning, refer to the coding of the ForTest instruments)
- four characters that, in hexadecimal format, indicate the active instrument optionals (for correspondence between number and meaning, refer to the coding of the ForTest instruments)
- four characters that, in hexadecimal format, indicate the active model optionals (for correspondence between number and meaning, refer to the coding of the ForTest instruments)

Section identifying some parameters coinciding with calibration parameters of the instrument, useful for better interpreting other instrument parameters

- two characters indicating the calibration unit of measure of the direct pressure (see Appendix 5)
- two characters indicating the calibration number of decimal points of the direct pressure
- two characters indicating the calibration unit of measure of the VOUT (see Appendix 5)
- two characters indicating the calibration number of decimal points of the VOUT
- two characters indicating the calibration unit of measure of the volume (see Appendix 5)
- two characters indicating the calibration number of decimal points of the volume
- two characters at zero
- two characters at zero
- two characters indicating the calibration unit of measure of the time (see Appendix 5)
- two characters indicating the calibration number of decimal points of the time

Section identifying some parameters useful for better setting some instrument parameters

- two characters indicating the setting unit of measure of the direct pressure (see Appendix 5)
- two characters indicating the setting number of decimal points of the direct pressure
- two characters indicating the setting unit of measure of the VOUT (see Appendix 5)
- two characters indicating the setting number of decimal points of the VOUT
- two characters indicating the setting unit of measure of the volume (see Appendix 5)
- two characters indicating the setting number of decimal points of the volume
- two characters at zero
- two characters at zero
- two characters indicating the setting unit of measure of the time (see Appendix 5)
- two characters indicating the setting number of decimal points of the time
- two characters indicating the difference of decimal points between a set pressure compared to calibration

- two characters indicating the difference of decimal points between the displayed pressures compared to calibration
- two characters at zero
- two characters at zero

Start index section in the various menus

- three characters indicating the position of the test menu of the first active parameter (see Appendix 4: it indicates all parameters of all instruments, but it may be that one instrument does not have all parameters active and, therefore, the first active parameter also may be different from one instrument to the other)
- three characters indicating the position of the setup menu of the first active parameter (the same previous consideration is valid)
- three characters indicating the position of the piece counter menu of the first active parameter (the same previous consideration is valid)
- three characters indicating the position of the version menu of the first active parameter (the same previous consideration is valid)
- three characters indicating the position of the calibration menu of the first active parameter (the same previous consideration is valid)
- three characters indicating the position of the submenu of the first active parameter (the same previous consideration is valid)
- three characters at zero
- three characters at zero
- five characters indicating the identification of the microcontroller

3.4 Command "4": Reset and/or Piece Counter Reading only

With regard to the command, the command character is "4" and the requested data characters are:

- one subcommand character, for the meaning of which refer to the following subparagraphs.

With regard to the response, it contains both the subcommands:

- again the subcommand, one character
- ten characters for the number of good
- ten characters for the number of rejects
- four characters for the year in which the last reset was executed (from "2000" to "2099")
- two characters for the month in which the last reset was executed (from "01" for January to "12" for December)
- two characters for the day in which the last reset was executed (from "01" to "31")
- two characters for the hour in which the last reset was executed (from "00" to "23")
- two characters for the minute in which the last reset was executed (from "00" to "59")

SUBCOMMAND "0": PIECE COUNTER READING

In this case the subcommand is "0". Therefore, if wanting to read the piece counter send:
":01403A".

SUBCOMMAND "1": RESET AND CONSEQUENT READ OF THE PIECE COUNTER

In this case the subcommand is "1". Therefore, if wanting to reset the piece counter send:
":014139".

NB: in this case, firstly reset then send the response. It will contain the good and reject piece count at "0", and the date and time of the last reset, coinciding with the "current" time, as the piece counter has just been reset.

3.5 Command “5”: Write the Program Number

With regard to the command, the command character is “5” and the requested data characters are:

- five characters indicating the number of the program to be set.

Setting the program number corresponds to loading the program number and all its parameters: successive start makes a test corresponding to this program number start. If wanting to load program “01”, send
“:0150000178”.

With regard to the response, it contains:

- the program number again, five characters

In case the program number passed to the instrument is incorrect (outside the admitted limits) or there is a test already in progress, the data field is filled with “e”.

3.6 Command “6”: Special Functions like Keys

With regard to the command, the command character is “6” and the requested data characters are:

- one subcommand character, for the meaning of which refer to the following subparagraphs.

With regard to the response, it contains for all subcommands:

- again the subcommand, one character

If the received subcommand is not correct, in place of the subcommand, “e” will be received (there are other cases where “e” is received in place of the subcommand, see individual description of the subcommands).

In all cases, the response is made of 7 characters.

3.6.1 SUBCOMMAND “1”: START THE TEST

In this case the subcommand is “1”. If wanting to START the test, send:

“:016137”.

If the test is already in progress, a new START will be ignored and will not affect operation of the test, and “e” will be received in the response in place of the start subcommand.

3.6.2 SUBCOMMAND “2”: ABORT THE TEST

In this case the subcommand is “2”. If wanting to ABORT the test, send:

“:016236”.

If the test is already in progress, a new ABORT will be ignored and will not affect operation of the test, and “e” will be received in the response in place of the abort subcommand.

3.6.3 SUBCOMMAND “3”: EXECUTE AUTOZERO

In this case the subcommand is “3”. Therefore, if wanting to execute an autozero, send:

“:016335”.

If the test is already in progress, an autozero will be ignored and will not affect operation of the test, and “e” will be received in the response in place of the start subcommand.

3.7 Command "7": Command "7": Read Barcode Command (optional)

With regard to the command, the command character is "7" and no data characters are requested. If wanting to read the buffer collecting the read barcode, send

"01767"

With regard to the response, it contains:

- three characters indicating the length of the read barcode
- forty characters indicating the read barcode.

In case the read barcode is excessively long, the characters indicating its length are equal to "e", like all characters indicating the rear and proper barcode. If the barcode is long n characters, then the first n of the forty characters in the response will be equal to the read barcode, whereas the last forty less n will be "e".

3.8 Command “8”:

Command not implemented.

3.9 Command “9”:

Command not implemented.

3.10 Command "A":

Command not implemented.

3.11 Command "B": Read a Menu Parameter

With regard to the command, the command character is "B" and the requested data characters are:

- two characters indicating the menu in which the parameter to be read is contained (see Appendix 4)
- two characters indicating the eventual submenu in which the parameter to be read is contained (see Appendix 4)
- five characters indicating the number of the program which parameter is to be read (if test menu, otherwise this field must be at zero for the other menus)
- three characters indicating the position in table of the parameter to be read (see Appendix 4)

If wanting to read parameter "10" of menu "02" (hypothesised not to be the setup menu, therefore the field of the program number goes to zero) submenu "00", send

":01B02000000001019".

With regard to the response, it contains:

- again the menu in which the parameter to be read is contained, two characters
- again the eventual submenu in which the parameter to be read is contained, two characters
- again the number of the program which parameter is to be read, five characters
- again the position in table of the parameter to be read, three characters
- five characters for the value of the parameter requested to be read (limited)
- two characters indicating the mark with which the read value must be interpreted: if worth "0" it is a positive parameter without mark (that increases from its minimum to maximum value, meaning if its limits allow it, it increases from 0 to 65535), if worth "1" it is a parameter with mark (that increases from its minimum to maximum value, meaning if its limits allow it, it increases from -32768 to 32768: in this case the read value must be interpreted, that a "logic" value -1 reaches serial as "65535"), if worth "2" it is a positive parameter with minus mark in front (at rate of case "0", but for reasons of customer side logic operation, it must be thought of as negative, meaning if "122" is received it is to be thought of as -122), if worth "3" it is a negative parameter (that increases from its minimum to maximum value, meaning if its limits allow it, it increases from -32768 to 0)
- five characters indicating the minimum value the parameter must assume (useful to know for successive writing)
- five characters indicating the maximum value the parameter must assume (useful to know for successive writing)
- five characters indicating the unit of measure of the read parameter (see Appendix 5)
- five characters indicating the number of decimal points associated to that parameter
- three characters indicating the next parameter available in that menu

In case the menu passed in the command is incorrect (outside the limits or not corresponding to any menu, see Appendix 4), the corresponding field is filled with "e". In case the submenu passed in the command is incorrect (outside the limits or not corresponding to any submenu, see Appendix 4), the corresponding field is filled with "e". In case the number of program passed to the instrument is incorrect (outside the admitted limits), in case the test menu is passed, or is different from zero in case a menu different to that of test is passed, the corresponding field is filled with "e". In case the number corresponding to the parameter position is incorrect (outside the limits or not corresponding to any parameter, see Appendix 4), the corresponding field is filled with "e". In all these cases, the successive fields described above will be full of "e" in the response.

To know whether a parameter is effectively active on the instrument (meaning the instrument is sensitive to this parameter), command "G" must be used to have a list of active parameters.

If from panel you are on the parameter being read through this command (the instrument display is displaying the parameter within this reading flashing), then the last memorised value and not that currently modified will be read in the response. To ensure a correct reading, carry out this command request when not on the parameter being read.

HOW TO READ AN ENTIRE MENU

It may be useful to have the possibility, in particular in a program from pc, to be able to read all

parameters relating to a menu.

For example, if wanting to read all parameters of the TST menu of an instrument, therefore all parameters relating to a program that may be program 1.

To start, it is necessary to know which is the first available parameter in this menu; to have this information send the version control to the instrument (See Control 3). Through the version control it is possible to know which is the address of the first active parameter, meaning the position in table of the first useful parameter.

By adequately composing the request string as explained above and sending it to the instrument, this will respond with the value of the parameter and the above indicated information and, also, with the position in table of the next useful parameter.

This information represents the next parameter to read and, if inserted in control, will give the possibility of requesting the next parameter.

It is easy to note that, with this management, all parameters available in any menu can be read.

To understand when the last parameter has been reached, verify that the next parameter is worth zero. If equal to zero, the instrument informs there are no other parameters available for a menu.

3.12 Command “C”: Write a Parameter of a menu

With regard to the command, the command character is “C” and the requested data characters are:

- two characters indicating the menu in which the parameter to be written is contained (see Appendix 4)
- two characters indicating the eventual submenu in which the parameter to be written is contained (see Appendix 4)
- five characters indicating the number of the program which parameter is to be written (if test menu, otherwise this field must be at zero for the other menus)
- three characters indicating the position in table of the parameter to be written (see Appendix 4)
- five characters containing the numerical value to be written in the parameter.

If wanting to write "10" in parameter "05" of menu "01" (supposed to be the test menu), submenu "00" of program "01", therefore send

“:01C0100000010050001023”.

With regard to the response, it contains:

- again the menu in which the parameter to be read is contained, two characters
- again the eventual submenu in which the parameter to be read is contained, two characters
- again the number of the program which parameter is to be read, five characters
- again the position in table of the parameter to be read, three characters
- five characters for the value effectively written in the parameter requested to be written: the instrument (with this control) adopts a control of the maximum and minimum limits for each parameter, in order to avoid errors by the programmer

In case the menu passed in the command is incorrect (outside the limits or not corresponding to any menu, see Appendix 4), the corresponding field is filled with “e”. In case the submenu passed in the command is incorrect (outside the limits or not corresponding to any submenu, see Appendix 4), the corresponding field is filled with “e”. In case the number of program passed to the instrument is incorrect (outside the admitted limits), in case the test menu is passed, or is different from zero in case a menu different to that of test is passed, the corresponding field is filled with “e”. In case the number corresponding to the parameter position is incorrect (outside the limits or not corresponding to any parameter, see Appendix 4), the corresponding field is filled with “e”. In case the numerical value to be written in the parameter is incorrect (outside of the general admitted limits 0 and 65535), the corresponding field is filled with “e”. In case any of these cases verifies, the instrument does not write anything. As said above, it is possible that the numerical control and response value be different between them: the response contains the value effectively written in the parameter.

To know whether a parameter is effectively active on the instrument (meaning the instrument is sensitive to this parameter), command “G” must be used to have a list of active parameters.

If from panel you are on the parameter being written through this command (the instrument display is displaying the parameter within this writing flashing), then the last effectively written value will be received in the response but, if memorisation of a different value is confirmed using the keys, then the writing through keys being the last performed, will win. To ensure a correct writing, carry out this command request when not on the parameter being written.

3.13 Command “D”:

Command not implemented.

3.14 Command “E”: Write a Memory Location

With regard to the command, the command character is “E” and the requested data characters are:

- two characters indicating the type of memory from which to read (see Appendix 7)
- four characters indicating the address in which to write (see Appendix 7), in hexadecimal format
- five characters indicating the number to write in the location

If wanting to write "50" in location number "20" of memory "02", send
":01E0100020005041".

With regard to the response, it contains:

- again the type of memory from where to read, two characters
- again the address in which to write, four hexadecimal characters
- again the number to be written in the location, five characters.

In case the type of memory passed to the instrument is incorrect (outside the limits or not corresponding to any memory, see Appendix 7), the corresponding field is filled with “e”. In case the starting address passed is incorrect (outside the limits or not corresponding to any address, see Appendix 7), the corresponding field is filled with “e”. In case the number is incorrect (outside the admitted limits, 0 and 65535), the corresponding field is filled with “e”. In case any of these cases verifies, the instrument does not write anything. It must be noted that, unlike command "C", in this case it is not possible to implement by the instrument an intelligent limit of the value that is written in the memory location.

Even if more difficult to understand, the same considerations as parameters writing are valid, meaning if from panel a parameter is being modified which address coincides with that to be written in this command, there may be future misunderstandings.

WARNING: the mapping includes variables at 8, at 16, at 32 bit, therefore before writing a variable, it may be necessary to read the same location (and if needed of successive), one (or more) writing masks and more writing commands may be required (see Appendix 7).

3.15 Command “F”: Set Date and Time

With regard to the command, the command character is “F” and the requested data characters are:

- two characters for the day
- two characters for the month
- four characters for the year
- two characters for the hour
- two characters for the minute
- two characters for the seconds.

If wanting to set the 29th October 2014 as date and 14:59:01 as time, send:

":01F2910201414590191".

With regard to the response, it contains:

- two characters for the day (limited between "01" to "31")
- two characters for the month (limited between "01" for January, "12" for December)
- four characters for the year (limited between "0000" to "9999", if year is 2009, it will be "2009")
- two characters for the time (limited between "00" to "23")
- two characters for the minute (limited between "00" to "59")
- two characters for the seconds (limited between "00" to "59")

It shows that upon writing, the instrument implements a certain limit to the values to be written. In case one of the sent values is outside of the admitted limits for the same, it will contain "e" in the response, and as not considered valid it will not be written in the memory (whereas all others yes). The instrument does not implement a control on leap years.

3.17 Command “G”:

Command not implemented.

3.17 Command “H”:

Command not implemented.

3.18 Command "T":

Command not implemented.

3.19 Command “J”:

Command not implemented.

3.20 Command “K”:

Exclusive internal use command.

3.21 Command “L”: Read Sensors

Exclusive internal use command.

3.22 Command “M”: Calibrate

Exclusive internal use command.

3.23 Command “N”: Read Calibration Points

Exclusive internal use command.

4. Protocol Communication Through TTY Description

The communication protocol, the commands, the master side settings are exactly the same as in case of direct communication.

The only difference in using the TTY communication (ring current) consists in a hardware - software enabling inside the instrument, that the instrument executes by adequately setting the setup parameter stated in paragraph 0.

5. Test End Track Operation Description

Communication with the instrument happens with the same settings for the communication with PC.

Like for communication towards PC, ASCII characters are sent: with regard to format, all addresses are passed in hexadecimal format, data in decimal, then both converted in ASCII characters; second and third characters of the communication strings are dedicated to the serial address of the instrument. This address is in hexadecimal format (e.g. address 1 will be inserted in the string as "01", address 30 will be inserted as "1E").

The instrument spontaneously sends test end string, and there is no possibility of request of any command by the PC.

The first sent character is always ":", follow two characters indicating the serial address of the instrument that sent the track, and for analogy to communication with PC (see paragraph 1) one "0" as command character. Therefore, the track starting with ":010" and ending with checksum (to calculate checksum see Appendix 1), contains the same response fields to result command (see paragraph 3.2), with the exception of the number of incorrect results, not having reason to exist here: in addition to not having the possibility of requesting incorrect results in case there are some, there cannot be, as when the instrument sends the test end track, it considers it always received, there not being any type of confirmation by the PC.

The response is therefore made of 109 characters.

Appendix 1: Checksum Calculation

The steps for calculating the checksum are the following:

1. Sum the ASCII value of all characters of the string, excluding the initial character ":" (ASCII value means, for example, for character "0" number "48", for character "A" number 65)
2. With the number in this way obtained, do I AND logic with 255.
3. Subtract the number obtained at point 2 from 255.
4. convert the number obtained at point 3 in hexadecimal

Below are the rows of code in Visual Basic 6.0 for calculating the checksum

For i% = 2 To lungs

vChk = vChk + Asc(Mid(Mtx\$, i%, 1))

Next i%

vChk = vChk And 255

vChk = 255 – vChk

Format\$(Hex\$(vChk), "00")

Appendix 2: Coding of the Test Phases

With regard the characters identifying the state of the test (or association number/meaning of the substatus of the test state in progress),

- "01": identifies the start
- "02": identifies the delay phase to start
- "03": identifies the pre-filling phase
- "04": identifies the pre-filling phase
- "05": identifies the pre-filling phase
- "06": identifies the pre-filling phase
- "07": identifies the pre-filling discharge phase
- "08": identifies the pre-filling discharge phase
- "10": identifies the pre-filling discharge phase
- "11": identifies the filling phase
- "12": identifies the filling phase
- "13": identifies the filling attempts charge phase
- "14": identifies the settling phase
- "15": identifies the settling phase
- "16": identifies the settling phase
- "17": identifies the settling phase
- "18": identifies the settling phase
- "19 – 25": internal usage only
- "26": identifies the test phase
- "27 – 29": internal usage only
- "30": identifies the patency test phase
- "31 – 35": internal usage only
- "36": identifies the seal test of the bell phase
- "37 – 42": internal usage only
- "43": identifies the volume test phase
- "44 – 49": internal usage only
- "50": identifies the end of the test phase with GOOD result.

In case of reading other values on the string send by instrument, ignore as exclusive internal use.

Appendix 3: Coding of Results

With regard to characters identifying the currently active result, below are the values they can assume with relative meaning:

- "0" identifies no active result
- "1" identifies as active result one good
- "2" identifies as active result one reject
- "3" identifies as active result one good with reserve (optional)
- "4" identifies as active result one reverse drop
- "5" identifies as active result one reject of the reference piece
- "6" identifies as active result one reject due to the bell
- "7" identifies as active result one reject for capacity under threshold (optional)
- "8" identifies as active result one reject for high full scale pressure
- "9" identifies as active result one reject for high full scale VOUT
- "10" identifies as active result one reject for pressure under tolerance
- "11" identifies as active result one reject for pressure above tolerance
- "12" identifies as active result one reject for pressure that has not reached or maintained the level
- "13" identifies as active result one abort
- "14" identifies as active result one reject for flow over the maximum set
- "15" identifies as active result one reject for minimum filling time
- "16" identifies as active result one reject for wrong volume measured
- "17" identifies as active result one reject for loss of charge low
- "18" identifies as active result one reject for loss of charge high
- "98" identifies as active result one abort for automations not carried out or interrupted, or that a pre-test automation phase (optional) is in progress
- "99" identifies that a test is in progress

Appendix 4: Index Coding and Tables for Reading and Writing Parameters

With regard the characters identifying the menu index, the menu number correspondence is the following:

- "1" corresponds to test menu
- "2" corresponds to setup menu
- "3" corresponds to piece counter menu
- "4" corresponds to version menu
- "5" corresponds to calibration menu.

Depending on the active menu, the correspondence between the characters identifying the parameter and the parameter itself is different meaning a different meaning corresponds to parameter number "3", depending on the menu it is in: see below.

With regard the characters identifying the submenu index, the submenu number correspondence, regardless of the menu, is the following:

- "1" corresponds to characters submenu.

The above menu fully correspond to those selectable in the instrument through relative keys, and the same parameters can be recalled through user and serial panel: therefore, for a more in-depth understanding on their use and interpretation, refer to the user manual provided with the instrument.

For each menu, below is a table containing the number associated to parameter for that menu in the first column, the meaning of the associated parameter in the next column (see user manual) and, finally, a column containing information on how to interpret the parameter itself (with regard to this column, relating to the decimal points and units of measure, this information is the same that can be deducted by simply observing the parameter itself on display; whereas, in case of parameters giving textual information on display, the number/text association must be understood). It can be noted how the increasing order of the parameter number faithfully follows the list order of the same parameters in the chapter of the relative menu of the manual. And, in case of parameters with textual meaning, the number/text association following an increase coinciding with what can be done with setting via keys.

TEST MENU

Number	Associated parameter	Interpretation (decimal points, unit of measure, coding)
003	Type of test	The parameter has the meaning of: "0"= decay, "1"= obstruction, "2"=functional cycle, "100"= flow test, "101"=slope Only for T8090 Dual Absolute® equipment the parameter has the meaning: "0" = Dual Absolute, "1"= Obstruction , "2"= Functional Cycles, "6" = Differential , "7"=Absolute, "9"=Zero-center
004	Filling pressure	Decimal points and units of measure obtainable from command 3 (pressure to be set).
005	Pressure tolerance	No decimal point, it is a percentage.
006	Filling type/attempts (see manual to understand when "filling type" and when "filling attempts")	If the parameter has the meaning of "Filling type" then "0" = time and "1" = level. If the parameter has the meaning of "Filling attempts" then the parameter is a number with zero decimal points and no unit of measure.
008	Filling time	Decimal points and units of measure obtainable from command 3 (time to be set).
009	Adjustment time	Decimal points and units of measure obtainable from command 3 (time to be set).
010	Test time	Decimal points and units of measure obtainable from command 3 (time to be set).
011	Discharge time	Decimal points and units of measure obtainable from command 3 (time to be set).
013	Minimum flow	Decimal points and units of measure obtainable from command 3

		(time to be set).
014	Maximum flow	Decimal points and units of measure obtainable from command 3 (time to be set).
015	Pressure>flow compensation / Volume of piece (see manual)	If the parameter has the meaning of "Pressure>flow compensation" then it has zero decimal points and it is a percentage. If the parameter has the meaning of "Volume of piece", decimal points and units of measure are obtainable from command 3 (volume to be set).
016	Mass>volume temperature	No decimal point, degrees centigrades.
018	Type of measurement	Set the type of measurement of the loss. If "Type of measurement": "0"= pressure, "1"= pressure/time, "2"= cc/h, "3"= cc/min.
019	Maximum drop	Decimal points and units of measure are obtainable from command 3 (VOUT to be set).
020	Volume of piece	Decimal points and units of measure are obtainable from command 3 (volume to be set).
021	Offset on measurement	Decimal points and units of measure are obtainable from command 3 (VOUT to be set).
023	Obstruction minimum pressure	Decimal points and units of measure are obtainable from command 3 (pressure to be set).
024	Obstruction maximum pressure	Decimal points and units of measure are obtainable from command 3 (pressure to be set).
028	Bell test time	Decimal points and units of measure obtainable from command 3 (time to be set).
029	Leak to be detected	Decimal points and units of measure are obtainable from command 3 (VOUT to be set).
030	Bell test frequency	No decimal point or unit of measure
033	Volumetric test time	Decimal points and units of measure obtainable from command 3 (time to be set).
034	Tolerance on volume	One decimal point, it is a percentage.
038	Pre-filling pressure	Decimal points and units of measure obtainable from command 3 (pressure to be set).
039	Pre-filling time	Decimal points and units of measure obtainable from command 3 (time to be set).
040	Pre-filling discharge time	Decimal points and units of measure obtainable from command 3 (time to be set).
041	Pre-filling Number/Type (see manual)	If the parameter has the meaning of "Pre-filling type" then "0" = time and "1" = level. If the parameter has the meaning of "Number of pre-fillings" then the parameter is a number with zero decimal points and no unit of measure.
043	Minimum filling time	Decimal points and units of measure obtainable from command 3 (time to be set).
044	Number of force measurement cycles	No decimal point or unit of measure.
045	Duration of force measurement cycle	Decimal points and units of measure obtainable from command 3 (time to be set).
048	Linked program	No decimal point or unit of measure.
049	Delay to start	Decimal points and units of measure obtainable from command 3 (time to be set).
050	Coupled output	No decimal point or unit of measure. Attention: it is a decimal parameter that must be interpreted as hexadecimal to understand which bit (output) is active or not.
051	Coupled input	No decimal point or unit of measure. Attention: it is a decimal parameter that must be interpreted as hexadecimal to understand

		which bit (input) is active or not.
053	Good percentage with reserve	No decimal point, it is a percentage.
058	Unit measure of the filling pressure	No decimal point or unit of measure.
059	Unit measure of VOUT (on a pressure decay instrument it is valid if VOUT is in pressure mode)	No decimal point or unit of measure.
060	Auxiliary parameter	No decimal point or unit of measure.

SETUP MENU

Number	Associated parameter	Interpretation (decimal points, unit of measure, coding)
002	Day	No decimal point or unit of measure.
003	Month	No decimal point or unit of measure.
004	Year	No decimal point or unit of measure.
005	Hour	No decimal point or unit of measure.
006	Minutes	No decimal point or unit of measure.
007	Maximum brightness	No decimal point, it is a percentage.
008	Minimum brightness	No decimal point, it is a percentage.
010	Serial address	No decimal point or unit of measure.
011	Type of start	"0"= bi-manual, "1"= panel excluded, "2"= from panel, "3"= 4 start, "4"= BCD+strobe, "5"= BCD+time.
012	Language	"0"= Italian, "1"= English, "2"= Spanish, "3"= French, "4"= German, "5"= Portuguese.
013	Chart enabling	"0"= off, "1"= thin, "2"= thick, "3"= thin dashes, "4"= thick dashes.
015	Label time	Decimal points and units of measure obtainable from command 3 (time to be set).
016	Acoustic signal time	Decimal points and units of measure obtainable from command 3 (time to be set).
017	Filling valve delay	Decimal points and units of measure obtainable from command 3 (time to be set).

018	Main valve delay	Decimal points and units of measure obtainable from command 3 (time to be set).
020	Pressure means number	No decimal point or unit of measure.
021	VOUT means number	No decimal point or unit of measure.
022	Direct pressure decimal points display factor	No decimal point or unit of measure.
025	Capacity scale threshold	No decimal point, it is a percentage.
026	Mass>volume temperature	No decimal point, degrees centigrades.
030	Reverse Drop	No decimal point, it is a percentage.
031	To reject	"0"= continues test, "1"= stops test, "2"= stops but calculates.
032	Equalisation adjustment	No decimal point, it is a percentage.
035	COM1 Mode	"0"= nothing, "1"= PC directly, "2"= PC through TTY, "3"= Tack, "4"= Barcode, "5"= Lola small label, "6"= Lola large label. ATTENTION: if this parameter is modified, communication may be jeopardised.
036	Baud rate COM1	"0"= 2400bps, "1"= 4800bps, "2"= 9600 bps, "3"= 19200bps, "4"= 38400bps, "5"= 57600bps, "6"= 112000bps. ATTENTION: if this parameter is modified, communication may be jeopardised.
037	COM2 Mode	See COM1 Mode.
038	Baud rate COM2	See Baud rate COM1.
045	Use analogical output 0	"0"= electronic regulator, "1"= nothing, "2"= proportional to pressure, "3"= proportional to VOUT.
046	Use analogical output 1	See Use analogical output 0.
050	Autozero every	No decimal point, unit of measure in minutes.
055	Second branch pressure	Decimal points and units of measure obtainable from command 3 (pressure to be set).
056	Third branch pressure	Decimal points and units of measure obtainable from command 3 (pressure to be set).
060	Test end signal	"0"= standard, "1"= busy, "2"= ready, "3"= impulse.
061	Reject management	"0"= standard, "1"= all automatic, "2"= validation, "3"= remains full.
062	Abort management	"0"= standard, "1"= all automatic, "2"= remains full.
063	Outputs Mode	"0"= standard, "1"= results, "2"= macroR, "3"= macroR&A.
065	Percentage on settling time	No decimal point or unit of measure.
066	Percentage on filling time	No decimal point or unit of measure.
067	Adjustment attempts	No decimal point or unit of measure.
080	Number of labels to be printed	No decimal point or unit of measure.
081	Print linked also	No decimal point or unit of measure.
082	Print rejects also	No decimal point or unit of measure.
083	Print abort also	No decimal point or unit of measure.
085	PID : Proportional Factor	No decimal point or unit of measure.
086	PID : Derivative Factor	No decimal point or unit of measure.
087	PID : Basic times	Decimal points and units of measure obtainable from command 3 (time to be set).

PIECE COUNTER MENU

There are no parameters in this menu to be read or written with commands B and C, as the piece counter can be read or reset with command 4.

Also remember that the memorised results displayed in the piece counter menu by the instrument are buffer, except compared to that which can be read with command 2.

VERSION MENU

Attention: this menu, and its parameters, are read only, therefore ensure not to attempt writing with command C.

Number	Associated parameter	Interpretation (decimal points, unit of measure, coding)
002	Instrument high full scale pressure	Corresponds to parameter in Calibration Menu, refer to the table therein.
003	Instrument high full scale VOUT	Corresponds to parameter in Calibration Menu, refer to the table therein.
004	Checksum boot	No decimal point or unit of measure. It is the same that can be read with command 3: however, unlike the latter, in this case it is read in decimal format (it must then be converted in hexadecimal format).
005	Checksum firmware total	No decimal point or unit of measure. It is the same that can be read with command 3: however, unlike the latter, in this case it is read in decimal format (it must then be converted in hexadecimal format).
006	Serial number (top part)	No decimal point or unit of measure. To compose the full serial number, multiply the top part by 65535 and sum the bottom part.
007	Serial number (bottom part)	No decimal point or unit of measure. To compose the full serial number, multiply the top part by 65535 and sum the bottom part.
009	Instrument pneumatic optional	Corresponds to parameter in Calibration Menu, refer to the table therein.
010	Instrument optional	Corresponds to parameter in Calibration Menu, refer to the table therein.
011	Model optional	Corresponds to parameter in Calibration Menu, refer to the table therein.

CALIBRATION MENU

Attention: the following parameters have been set by ForTest for the correct operation of the instrument and, therefore, it must ever be necessary to modify them, whereas it is possible to read them. Some univocally define the instrument code, therefore, their modification jeopardises the identification of the instrument during after-sales assistance.

Attention: the parameters from 103 to 139 must not be read or written using commands B and C, as they correspond to the calibration of the various transducers assembled on the instrument. For calibration and reading of the calibrated points there are commands L, M and N.

Number	Associated parameter	Interpretation (decimal points, unit of measure, coding)
141	Instrument high full scale pressure	Decimal points and units of measure obtainable from command 3 (pressure to be set). This parameter is used for the instrument code.
142	Instrument high full scale VOUT	If it is an Mx99x or Mx97x instrument, the decimal points and units of measure are obtainable from command 3 (pressure to be set). For other instruments, decimal points and units of measure obtainable from command 3 (pressure to be set). This parameter is used for the instrument code.
143	Pressure module	With this parameter it is possible to read/write a numerical value to which another numerical value is associated: "0"= 1, "1"= 5, "2"= 10, "3"= 50, "4"= 100, "5"= 500, "6"= 1000, "7"= 5000.
144	VOUT module	See Pressure module.
146	Maximum pressure decimal points	No decimal point or unit of measure. Calibration parameter that can also be read with command 3. This parameter is used for the instrument code.
147	VOUT decimal points	No decimal point or unit of measure. Calibration parameter that can also be read with command 3. This parameter is used for the instrument code.
148	Pressure unit of measure	Number to be interpreted as textual. See Appendix 5 for the number/meaning association. Calibration parameter that can also be read with command 3. This parameter is used for the instrument code.
149	VOUT unit of measurement	Number to be interpreted as textual. See Appendix 5 for the number/meaning association. Calibration parameter that can also be read with command 3. This parameter is used for the instrument code.
151	AD Pressure gain	No decimal point or unit of measure.
152	AD VOUT gain	No decimal point or unit of measure.
153	AD Pressure speed	No decimal point or unit of measure.
154	AD VOUT speed	No decimal point or unit of measure.
156	Calibration temperature	No decimal point, unit of measure in degrees centigrades.
157	PLL Activation	"0"= no, "1"= yes. Attention: if this parameter is modified, switch the instrument off and on for the serial communication to start again (re-setting of the communication speed).
161	Instrument optional	No decimal point or unit of measure. This parameter is read/written in decimal format, but must be converted in hexadecimal format for its correct interpretation. This

		parameter is used for the instrument code.
162	Model optional	No decimal point or unit of measure. This parameter is read/written in decimal format, but must be converted in hexadecimal format for its correct interpretation. This parameter is used for the instrument code.
166	Volume decimal points	No decimal point or unit of measure. Calibration parameter that can also be read with command 3.
167	Volume unit of measure	Number to be interpreted as textual. See Appendix 5 for the number/meaning association. Calibration parameter that can also be read with command 3.
170	Dead volume	Decimal points and units of measure are obtainable from command 3 (volume to be set).
171	Reference volume	Decimal points and units of measure are obtainable from command 3 (volume to be set).
181	Pneumatic optional	No decimal point or unit of measure. This parameter is read/written in decimal format, but must be converted in hexadecimal format for its correct interpretation. This parameter is used for the instrument code.
182	Power supply voltage	The association number/meaning is the same as that used for coding of the ForTest products. This parameter is used for the instrument code.
183	Fittings	The association number/meaning is the same as that used for coding of the ForTest products. This parameter is used for the instrument code.
184	Type of gas	The association number/meaning is the same as that used for coding of the ForTest products. This parameter is used for the instrument code.

Appendix 5: Coding of the Unit of Measure

With regard to coding of the pressure unit of measure, below is a table for the association between numerical value and its meaning:

Number	Associated value	Size
0	mbar	Pressure
1	bar	Pressure
2	hPa	Pressure
3	Pa	Pressure
4	psi	Pressure
20	mbar/s	Pressure/time
21	bar/s	Pressure/time
22	hPa/s	Pressure/time
23	Pa/s	Pressure/time
24	psi/s	Pressure/time
40	cc/h	Volume/time
41	cc/min	Volume/time
42	l/h	Volume/time
43	l/min	Volume/time
60	s	Time
61	min	Time
70	cc	Volume
71	l	Volume
80	--	General
81	%	General
82	bps	Communication speed
83	°C	Temperature
84	conv/s	Conversion speed
85	prg	General
86	chin	General
87	chout	General
88	Volt	Electric voltage

Appendix 6: Coding of the I/O Masks

It must firstly be specified that in the response string of this instrument these masks have passed in decimal format, therefore, between 0 and 255 (being always 8 bit to be coded, therefore between 00 and \$FF). For example, if 97 (\$61) is read on the outputs mask it means the outputs corresponding to bit 6, 5 and 1 are active (supposing bit=1 means active output); if the inputs mask contains value 23 (\$E9) it means the inputs corresponding to bit 4, 2 and 1 are active (supposing bit=0 means active input).

Also, depending on the instrument, the association bit and connected input or output is different, and the correspondence on the same instrument may change depending on the enabled optionals. Therefore, below reference is made for each instrument to the basic configuration; to have all correspondences it is necessary to ask ForTest the characteristic data of a specific instrument.

For a better understanding of the use and operation of these I/O refer to the instrument manual.

M2xxx

bit	Active at	mask	Use
7..0	0	Inputs	
7..0	1	Outputs	
7..0	-	Expansion	Not envisioned

M79xxx

bit	Active at	mask	Use
7	0	Inputs	Not envisioned
6	0	Inputs	Not envisioned
5	0	Inputs	Start / Abort
4..0	-	Inputs	Not envisioned
7..4	-	Outputs	Not envisioned
3	1	Outputs	Electrovalve
2	1	Outputs	Electrovalve / Not envisioned
1	1	Outputs	STSA
0	1	Outputs	STSB
7..0	-	Expansion	Not envisioned

M77xxx

bit	Active at	mask	Use
7	0	Inputs	Input B
6	0	Inputs	Input A
5	0	Inputs	Start / Abort
4..0	-	Inputs	Not envisioned
7..4	-	Outputs	Not envisioned
3	1	Outputs	STSA
2	1	Outputs	STSB
1	1	Outputs	Electrovalve
0	1	Outputs	Electrovalve / Not envisioned
7..0	-	Expansion	Not envisioned

M6xxx

bit	Active at	mask	Use
7	0	Inputs	Outputs state diagnosis
6	-	Inputs	Used for electric power supply
5	0	Inputs	Start1 / BCD0 / Bi-manual

4	0	Inputs	Start
3	0	Inputs	Start3 / BCD2
2	-	Inputs	Used as output
1	0	Inputs	Start2 / BCD1
0	0	Inputs	Abort
7	1	Outputs	Electrovalve
6	1	Outputs	Cycle end signal
5	1	Outputs	Reject signal
4	1	Outputs	Good signal
3	1	Outputs	Electrovalve
2	1	Outputs	Discharge electrovalve
1	1	Outputs	Electrovalve
0	1	Outputs	Good signal with reserve (optional)
7..0	-	Expansion	Not envisioned

M8xxx, M9xxx

bit	Active at	mask	Use
7	0	Inputs	Outputs state diagnosis
6	0	Inputs	Start4 / BCD3
5	0	Inputs	Start1 / BCD0 / Bi-manual
4	0	Inputs	Start
3	0	Inputs	Start3 / BCD2
2	-	Inputs	Used as output
1	0	Inputs	Start2 / BCD1
0	0	Inputs	Abort
7	1	Outputs	Electrovalve
6	1	Outputs	Cycle end signal
5	1	Outputs	Reject signal
4	1	Outputs	Good signal
3	1	Outputs	Electrovalve
2	1	Outputs	Discharge electrovalve
1	1	Outputs	Electrovalve
0	1	Outputs	Good signal with reserve (optional)
7	0	Expansion	ING3 Expansion input
6	0	Expansion	ING2 Expansion input
5	0	Expansion	ING1 Expansion input
4	0	Expansion	ING0 Expansion input
3	1	Expansion	Electrovalve / OUT3 Expansion output
2	1	Expansion	Electrovalve / OUT2 Expansion output
1	1	Expansion	Electrovalve / OUT1 Expansion output
0	1	Expansion	Electrovalve / OUT0 Expansion output

Appendix 7: Memories and Mapping of the same

With regard the characters identifying the menu index, the menu number correspondence is the following:

- "1" corresponds to EEPROM memory
- "2" corresponds to RAM memory
- "3" corresponds to FM memory.

Depending on the active memory, the correspondence between the characters identifying the memory address and the corresponding parameter is different, meaning to address number "0" corresponds a different, depending on the memory it is in: see below.

For each memory, below is a table containing the address in hexadecimal format to be sent on serial in the first column, the associated parameter in the next column (see user manual) and, finally, a column for any notes; for information on how to interpret said parameter (decimal points and units of measure, refer to that said and explained in Appendix 4).

It is advised to use commands A, B and C rather than commands D and E, because easier at comprehension level and because they read and write in an intelligent manner, which allows not to influence the operation of the instrument. By directly writing in the memory there is the risk, in case the user mistakes the address or sets data outside of the admitted limits, to have anomalous behaviours of the instrument (which might not be clearly visible, but be present anyhow).

EEPROM

Address	Associated parameter	Notes

RAM

Address	Associated parameter	Notes

FM

Address	Associated parameter	Notes

Appendix 8 : Linked tests codification

For what concern the codification the program linking in the test result string, the field has the significance below:

Value	Meaning
0	No linked program
L	It's means that it there will be a program linked
E	End of the linked program sequence
S	Stop of a linked program sequence (ex. For a Reject or an Abort)

Changelog

03/06/2011 : Release 006 of the serial protocol manual.

17/06/2011 : Release 006.1; status command corrected. The first series of characters, that indicates the state of errors in the instrument, is composed by 4 characters and not of 5 characters.

07/10/2011 : Release 006.2; paragraph enumeration corrected.

11/01/2012 : Release 006.3;

-) Instantaneous command status : flag errors in hexadecimal format.

-) Added appendix nr 8; linked program explanation for the result string

-) Version command corrected. Reverted VOUT field with second character “-“

-) Appendix nr. 4 correction; 024 address, max pressure obstruction test

-) Add test phases in Appendix nr. 2

-) Command “B” correction; print error in the first example.

17/09/2013 : Release 006.4;

-) Added in the appendix 4 (TEST MENU) the addresses of the unit measure of pressure and vout

26/03/2014 : Release 006.5;

-) Corrected read counter command. Corrected the checksum value in the reset counter example

11/04/2014 : Release 006.6

-) Correct appendix 4; parameter 3 (kind of test)

29/10/2014 : Release 006.7

-) Correct the write command for datetime

07/06/2018 : Release 006.8

-) Update on appendix 3 with codes upper to 13